Inequalities — problem 7.9

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Let $a, b, c, d > 0$. Find all possible values of the sum

$$S = \frac{a}{a+b+d} + \frac{b}{a+b+c} + \frac{c}{b+c+d} + \frac{d}{a+c+d} \qquad \text{(IMO 1974)}.$$